

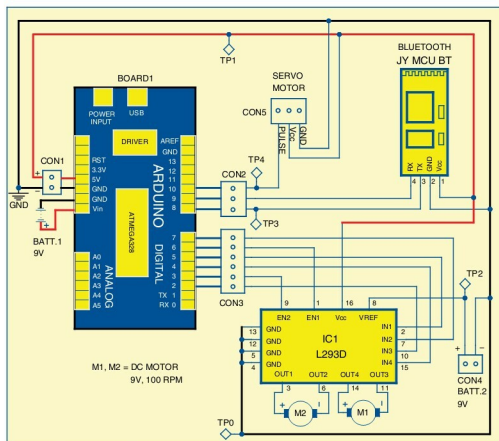


Fig. 3: Circuit of the soccer robot

every 20 milliseconds. The length of the pulse will determine how far the motor turns. For example, a 1.5-millisecond pulse could turn the motor to 90-degree position (often called the neutral position). If the pulse is shorter than 1.5 milliseconds, the motor would turn lesser. If the pulse is longer than 1.5 milliseconds, the shaft would turn closer to 180 degrees as

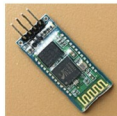


Fig. 4: Bluetooth module

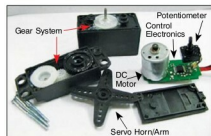


Fig. 5: Servo motor components

shown in Fig. 6.

Servo motor arm is connected to a metal or wooden piece, so that the servo arm acts as a leg to kick the ball.

Arduino UNO board. Arduino is

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	9V
TP3	Train of pulses
TP4	PCM pulses for servo motor control

PARTS LIST

Semiconductors:	- L293D, motor driver
IC1	
Miscellaneous:	- Arduino UNO board
BOARD1	- JY MCU BT, Bluetooth module
Bluetooth	
Servo motor:	- 5V Vigor VS-2 servo motor
CON1, CON4	- 2-pin connector
CON2, CON5	- 3-pin connector
CON3	- 6-pin connector
M1, M2	- 9V, 100rpm DC motor
BATT.1, BATT.2	- 9V battery

an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino UNO is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an in-circuit serial programming (ICSP) header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable, or power it with an AC-to-DC adapter or battery, to get it started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

Pins 8 and 9 of BOARD1 are connected to pins Tx and Rx of the Bluetooth module, respectively. Pin 10 is connected to PULSE pin of servo motor to control the angle of movement. Pins 2 through 7 of BOARD1 are output pins that are connected to IC1 for controlling the motors. Pins 2, 3 and 4 of BOARD1 are connected to IN3, EN2 and IN4 of IC1 to control motor M1, and pins 5, 6 and 7 are connected to IN1, EN1 and IN2 of IC1 to control motor M2. EN1 and EN2 are used to control speeds of the motors.

The app sends the sensed data from accelerometer of the phone to the robot via Bluetooth. Data transmitted by the phone is received by the Bluetooth module on the robot. The received data is fed to pin 8 of BOARD1. The microcontroller on BOARD1 processes the received data and drives